Programming Fundamentals II: Assignment 02

Due Date: Saturday October 26, 2024 at Midnight

Library Platform

For this assignment, we are going to design several classes to emulate a small library system platform. You must determine which classes are parents or children, and which are enums, or abstract.

Our library will be lending out different types of items. We have 3 types of items in our system: Books, movies, and magazines. An item must be one of these three types.

- Every item in our library has a library identification number, a title, and a genre
- The genres our library offers are fantasy, sci-fi, history, horror, romance, and science. An item's genre must be one of these
- Books and magazines have publishers and page counts
- Each magazine has an edition, which is the month/year it was published
- Books have authors
- Movies have directors, studios, and runtimes
- Each book has a format, which is one of softcover, hardcover, or e-book
- Each item must have a getDescription method
  - Book descriptions are "*title* by *author* (*format*)" *e.g. "The Hobbit by Tolkien (HARDCOVER)"*
  - Movie descriptions are "*title* directed by *director*" *e.g. "The Hobbit directed by Jackson"*
  - Magazine descriptions are "*title (edition)"* *e.g. "National Geographic (Oct 2024)"*
- The toString of an item should return that item's description

a. **(16 Marks) Hierarchy Design**
   Code all of the required classes to support the description of the system above. Your design should allow us to store any items in a single array/ArrayList based on what type of product. Include only full-arg constructors. You may leave out any getters and setters that are not used.
b. **(6 Marks) Library Class**
   Create a Library with the following features:
   - The library class must manage an ArrayList of library items, which can hold magazines, books, or movies
   - A library has one constructor that accepts the name of the library, and initializes an item list of items
   - Include an addItem() method which will add a library item to the ArrayList
   - Include a checkout() method which will accept the title of an item and return that item. This method should remove the item from the ArrayList
   - If the title cannot be found in the ArrayList, throw an ItemNotFoundException
   - ItemNotFoundException should be a custom runtime exception that you created

There is no driver class included with this assignment. Feel free to create your own, but it is not for marks.

Add the **date and your name and student number** to the top of your java files. Please submit your Java files to Nexus before the end of the day on Saturday October 26 at Midnight. **Late submissions will be not be accepted.**